



# UNIVERSITY OF CENTRAL PUNJAB LAHORE

## Assignment # 1 CLO Mapping: CLO1

|                       |                                           |                 |            |
|-----------------------|-------------------------------------------|-----------------|------------|
| Course Code           | MAT233                                    | Semester        | Fall 2025  |
| Course Title          | Linear Algebra                            |                 |            |
| Resource Person       | Dr. Muhammad Rayees Ahmad                 |                 |            |
| Assignment Given Date | 22-10-2025                                | Submission Date | 27-10-2025 |
| Total Marks           | 30= 25 (Assignment) + 5 (Quiz-based Viva) |                 |            |

| Name of student | Complete Registration no. | Section | Assignment marks | Viva marks | Total marks |
|-----------------|---------------------------|---------|------------------|------------|-------------|
|                 |                           |         |                  |            |             |

### Submission Instructions (Please follow strictly)

- Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the questions on assignment papers or A4 papers. DO NOT use any other papers.
- This is an **INDIVIDUAL ASSIGNMENT**.
- It is mandatory to know the method and procedure of solving all questions. Any question may be asked in the viva.
- Write your complete name and registration number on front page of the assignment.
- Follow the deadline. Finish your work one day before, so you could submit in time.
- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).**
- All Plagiarized assignment will be awarded **ZERO** marks.

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student signature: \_\_\_\_\_

**Name:**

**Registration #:**

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**Q # 1 [CLO 1] (10 marks):**

Using **GeoGebra**, **Describe** and visualize the three possible types of solutions for a system of linear equations in three variables: **unique solution**, **infinite solutions**, and **no solution**. For each case, provide a **supporting example (other than lecture notes example)** of a system of equations and **explain** the geometric interpretation (point, line, or parallel planes). Attach the prints of GeoGebra working.

**NOTE: Each student's example must be different, otherwise ZERO marks will be given to those who has same examples.**

**Solution:**

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**Name:**

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**Q # 2 [CLO 1] (8 marks):**

In a cloud data center, four types of computing resources are being managed:

- **x**: number of CPU cores assigned per client
- **y**: amount of RAM (in GB) allocated
- **z**: amount of storage (in TB) provided
- **w**: network bandwidth units assigned

To balance performance and cost across different service levels, the system administrator sets up the following relationships between these resources for four client configurations:

$$\begin{cases} 3x - 4y - 17z + 13w = 11 \\ x - 2y - 7z + 7w = 14 \\ -x + 2y + 8z - 5w = 17 \\ 2x - 2y - 11z + 8w = 18 \end{cases}$$

- a) Each equation represents a **resource constraint** for a specific service plan. Your task is to find whether this system has a **unique**, **infinite**, or **no solution**, and interpret what that means for resource allocation (e.g., whether there is a consistent set of resource values that satisfies all service plans).
- b) **Solve** the system of linear equations by using Gauss-Jordan elimination method.

**Solution:**

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**Q # 3 [CLO 1] (7 marks):**

**Find** the values of  $a$  for which the system has no solutions, exactly one solution, or infinitely many solutions

$$x + y + 7z = -7$$

$$2x + 3y + 17z = -16$$

$$x + 2y + (a^2 + 1)z = 3a$$

Assume any 3 appropriate values of  $a$  to plot, using GeoGebra, corresponding planes which represent no solution, exactly one solution and infinitely many solutions. Attach the prints of the plots.

**Solution:**

**Name:**

**Registration #:**

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**Name:**

**Registration #:**

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